

Ankita Das

Curriculum Vitae

SRM University AP, Amaravati
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Research interest

My research interest widely lies in algebraic number theory. Currently I am interested in Jacobian varieties of super elliptic curves. Also I like to explore different aspects of algebraic geometry and the interplay between geometry and number theory.

Academic information

- 2024-Present **PhD in Mathematics**, SRM University, AP, and Indian Institute of Science Education and Research(IISER),Thiruvananthapuram
- 2020-2022 **Master of Science (M.Sc.) in Mathematics**, Jadavpur University
- 2017-2020 **Bachelor of Science (B.Sc.) in Mathematics**, Jadavpur University
- 2017 **Higher Secondary Examination (10+2 level)**, West Bengal Board, Sunity Academy
Percentage – 93.6
- 2015 **Secondary Examination (10 level)**, West Bengal Board, Sunity Academy
Percentage – 95.6

Papers published

- 2026 **Generalized Fruit Diophantine Equation And Super Elliptic Curves**, Published in Monatshefte Fur Mathematik, <https://doi.org/10.1007/s00605-026-02201-6>, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty

Projects

- Ongoing work in PhD **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, with Prof. Kalyan Banerjee and Prof. Kalyan Chakraborty
- Master's Project **A Survey on Clean Ring and Its Structural Behaviour**, link: https://drive.google.com/file/d/1QEc2nNCNo7x_4B1HTCkSsJS3VSQXVSbH/view?usp=drivesdk, under supervision of Prof Sukhendu Kar
- Reading Project **Invariant Theory of $n \times n$ matrices**, under supervision of Dr. Sarbeswar Pal

Some slides of my talks

- **On Jacobian on a Family of Superelliptic Curves**, link: https://drive.google.com/file/d/1lvwSfrl7jE-k-ZNXUjwckJ3xArooIjCm/view?usp=drive_link
- **Solvability problems in *Diophantine equations* via the arithmetic of elliptic and superelliptic curves**, link: <https://drive.google.com/file/d/1nq56pQEbzHCpuFzFdTPHH0ibFv4wzvHf/view?usp=drivesdk>
- **From Fruits to Class Groups**, link: https://drive.google.com/file/d/16ayXCVwTSfUipIeFV88a9r4S6PWTUVU_Q/view?usp=drivesdk
- **Factorization of Sine and Cosine functions and Proof of Walli's Formula**, link: https://drive.google.com/file/d/1EK-sX0JOREoSGXfnHoqfGao__CXsdR4g/view?usp=drive_link
- **Zariski-Nagata Theorem: A Few Consequences**, link: https://drive.google.com/file/d/1mKtPXQp_n2EJIIdaNx6fh-ftqd4r_NaHI/view?usp=sharing

Courses Taken in PhD Coursework

- | | |
|---|---|
| <ul style="list-style-type: none"> ○ Fourier Analysis ○ Advanced Complex Analysis ○ Advanced Measure Theory ○ Commutative Algebra ○ Differential Geometry ○ Manifold ○ Algebraic Number Theory ○ Algebraic Geometry | <ul style="list-style-type: none"> ○ Category Theory ○ Representation Theory ○ Infinite Galois Extension ○ Cohomological Algebra ○ Research Methodology ○ Research Publication and Ethics ○ Research Seminar |
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Courses Taken in Graduate Studies

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|---|---|
| <ul style="list-style-type: none"> ○ Abstract Algebra ○ Real Analysis ○ Topology ○ Linear Algebra ○ General Mechanics ○ Differential Geometry, Probability and Stochastic Process ○ Linear Algebra and Module Theory ○ Complex Analysis and PDE ○ Functional Analysis ○ Ordinary Differential Equations and Special functions ○ Advanced Topology ○ Dynamical System ○ Integral Equations, Integral Trans- | <ul style="list-style-type: none"> forms and Calculus of Variations ○ Algebraic and Differential Topology ○ Advanced Rings and Modules-I ○ Theory of Semigroups-I ○ Optimization and Control Theory ○ Advanced Functional Analysis ○ Advanced Rings and Modules-II ○ Theory of Semigroups-II ○ Topics in Analysis ○ Topics in Linear Algebra ○ Topics in Manifold ○ Commutative Algebra ○ Topics in Algebra ○ Differential Geometry |
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Courses Taken in Undergraduate Studies

- Algebra, Geometry and Calculus
- Abstract Algebra
- Statistics
- Real Analysis
- Ordinary Differential Equations
- Computer Science
- Linear Algebra
- Numerical Methods
- Partial Differential Equations
- Metric Space and Complex Analysis
- Linear Programming and Game Theory
- Probability
- Mechanics
- Number Theory
- Combinatorics and Graph Theory

Exams Qualified

- 2023 **Qualified Joint CSIR-UGC NET**
- 2023 **Qualified West Bengal State Level Eligibility Test, (WBSET)**
- 2021 **Qualified IIT JAM examination**
- 2017 **Obtained 94% marks in Mathematics, 90% marks in Physics in 10+2 standard.**
- 2015 **Obtained 100% marks in both Mathematics and Physical Science in 10 standard.**

Awards and Scholarships

- 2025 **President Scholastic Award, as "*Maintenance Allowance*" for securing *external government funding*, by SRM University, AP**
- 2023-Present **Joint CSIR-UGC JRF Fellowship, for qualifying all India *Joint CSIR-UGC NET*, by UGC**
- 2022-2021 **Indira Gandhi Single Girl Child Scholarship, by UGC**
- 2022-2017 **Bigyani Kanya Medha Britti (BKMB) scholarship, for securing position in *top 50 female candidates* in a state level exam organized by *JBNSTS***
- 2017 **DST-INSPIRE Scholarship, for securing position in *top 1 percent candidates* in 10+2 level state board examination, by *Scholarship for Higher Education (SHE)***

Work Experience

- Odd Semester
2025-2026 **TA for grading, CSE 203 (*Discrete Mathematics*), SRM University, AP**
- 2022 **Gramin Dak Sevak, Assistant Branch Post Manager, *Indian Sub Post Office*, Nilkuthi, Cooch Behar**

Technical Skills

- Programming Languages **LaTeX, C, C++, Python**

Conferences and Symposiums

July, 2026	41st Annual Conference of Ramanujan Mathematical Society (RMS), India
December, 2025	Indian Women and Mathematics (IWM), Research in Groups programme, India
December, 2025	Sakura Science Exchange Program, Japan
October, 2025	International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025, SRM University, AP, Andhra Pradesh, India
January, 2025	Frontier Symposium in Mathematics (FSM)-2025, IISER Thiruvananthapuram, Kerala, India
October, 2023	International Conference on Class Groups Of Number Fields And Related Topics (ICCGNFRT)-2025, Kerala School of Mathematics (KSoM), Kerala, India

Extra Curricular Activities

- I was trained in **Hindustani Classical Music** and **Rabindrasangeet** for several years. I like to listen various forms of semi-classical and light music. I enjoy different classical instruments. I won several district and state level music competitions.
- I was trained in **recitation** from Bangiya Sangeet Parishad and also completed my **Visharad** in it. I won various state and national level recitation competitions. I was in **AIR** as child recitation artist.
- I won a national essay writing competition on "**Eradication of Poverty**" organised by **TATA building India** and felicitated by **President of India** in 2014.
- Writing poetry, stories and reading books give me immense pleasure.
- I always enjoy to take part in instantaneous debate and won many district level competitions.

Languages

English, Hindi, Bengali